DiffOSeg: Consensus and Preference

CISC8006 · Claim reproduction and controlled extensions

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CISC8006 Advanced Machine Learning

2026-11-08

Does probabilistic consensus outperform averaging?

Stage I · LIDC-IDRI · one patient fold · three paired seeds

- GED30 should decrease
- DiceSoft30 should increase
- Both paired 95% intervals must support the expected directions

Observed outcome: inconclusive

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1.2 Consensus strategies

P: probabilistic consensus

Sample 1–4 annotators, choose a subset uniformly and average its masks.

A: average aggregation

Average all four annotation masks.

Both conditions share architecture, training budget, augmentation and paired initialization.

The frozen implementation samples a categorical target from the consensus probabilities.

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1.3 Frozen comparison

Item	Setting
Held-out data	54 patients, 450 patches
Training data	160 patients
Paired seeds	20261, 20262, 20263
Training	120000 updates each, batch 12
Diffusion / evaluation	250 cosine steps / 30 samples
Recorded compute	12.157 GPU-hours on one RTX 5090

Local protocol freeze precedes this P/A comparison. Earlier P-only runs had examined these patients.

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1.4 Three paired seeds

Seed	P GED30	A GED30	P DS30 (%)	A DS30 (%)
20261	0.103356	0.101623	90.3906	90.3736
20262	0.104700	0.107732	90.3073	90.0645
20263	0.104773	0.103719	90.2438	90.1576
Mean	0.104276	0.104358	90.3139	90.1986

P's GED improves in one seed and worsens in two.

DiceSoft has the expected ordering in all three seeds.

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1.5 Paired uncertainty

P minus A	Mean change	Paired 95% interval
GED30	−0.000082	[−0.005397, +0.005440]
DiceSoft30	+0.1153 pp	[−0.2292, +0.4960] pp

Both intervals include zero.

10,000 paired seed-by-patient bootstrap draws retain every patch of each sampled patient.

Patient-only and alternative empty-mask analyses reach the same outcome.

1.6 Discrepancy diagnosis

- Pairing, checkpoint identity and metric recomputation passed
- The local P/A gap is smaller than the paper's gap
- Subset sampling, categorical encoding and augmentation include disclosed assumptions
- Split composition and three-seed variability remain possible explanations

P and A have the same expected per-pixel consensus under the frozen sampler. Shared subsets can still change spatial dependence.

No controlled study isolates the cause of the discrepancy.

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2.1 Improvement portfolio

Hypothesis	Mechanism	Evidence
H1 Boundary	Correct localization	Full v1 + image-only control
H2 Polar	Model boundary variation	Full v2 negative result
H3 Votes	Supervise local disagreement	Full v3 + control
H4 Energy	Calibrate sample distribution	Failed pilot + full v4

These studies use the separately frozen E11 four-fold baseline.

Historical primary endpoints: GED50, DiceSoft50 and aligned preference Dice.

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2.2 Four-fold consensus results

Method	GED30	DS30 (%)	GED50	DS50 (%)
E11	0.119981	89.362	0.116899	89.510
Boundary v1	0.115828	90.077	0.113739	90.136
Polar v2	0.118440	89.057	0.116139	89.198
Votes v3	0.108311	90.711	0.107941	90.696
Energy v4	0.104974	90.771	0.104515	90.790

v4 versus v3: GED50 change −0.003426, CI [−0.005314, −0.001551].

Small positive Dice changes have intervals that cross zero.

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2.3 Energy calibration controls

Variant	GED50	DS50 (%)	Preference (%)
Full v4	0.104515	90.790	89.146
Location only	0.107806	90.692	89.144
No location	0.104752	90.789	89.155
Original sample order	0.103678	90.790	89.146
No energy readout	0.115012	90.730	89.146
Energy-only on v3	0.097209	90.483	89.137

Removing the energy readout worsens GED.

The original sample order beats the new ordering in this fitted system.

Controls remove components at inference without retraining.

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2.4 Supplementary preference result

Fold	Patients	Patches	E11 (%)	A (%)	E (%)
0	54	450	89.2673	90.2783	90.4602
1	54	375	84.9780	90.6664	90.7133
2	53	412	89.2580	90.5217	90.5803
3	53	372	88.5866	90.3461	90.3301
Mean	214	1609	88.0225	90.4531	90.5210

E: **90.5210%**, +2.4985 pp versus E11.

Deterministic CT guidance and three SAM2.1 seeds add models and compute. The historical 91% target remains unmet.

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3.1 Interpretation limits

- Original claim and extensions use different frozen baselines
- Repeated use of the cohort limits generalization claims
- Patient intervals condition on the fitted extension models
- Three ensemble members are not three full-system repetitions
- Complete historical compute accounting remains unavailable

Final E's preference Dice does not replace the P/A GED30 and DiceSoft30 test.

3.2 Evidence and team roles

Portable CPU review

```
python3 -B scripts/check_course_project.py
```

Full replay: 2,700 case groups and 81,000 predictions. Frozen-precision metric error: zero.

Codex technical review: recovered the P/A archive and verified the metrics and bootstrap.

Responsibility weights: Zheng 30%, Mu 30%, Liu 40%.

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3.3 Conclusions

The submitted P/A experiment is complete and inconclusive.

Energy calibration improves GED in the separate four-fold extension.

Negative studies and component controls constrain the explanation.

The evidence package makes the results inspectable while preserving their limits.

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